SPIKING NEURAL NETWORKS · SPIKING HEIDELBERG DIGITS

What a neuron's internal state buys

A vector-valued spiking neuron on a standard event-based benchmark

Mehran Jaberinezhad — Project overview · November 2026

88.07%
official TEST · 1,994 / 2,264
100 neurons · D = 1,000 · 3,069,201 parameters
30 epochs · seed 0 · FIT 99.78%

METHODS · WHAT THE MODEL IS

01 The question

A scalar spiking neuron carries one number. Can making that number a vector buy computation — and does the gain survive a standard benchmark?

$$\tau\, dV/dt = -V + I(t), \quad s_t = H(V_t - \theta)$$

The scalar abstraction is the default in event-based networks. We keep the spike, the leak, the threshold and the count readout, and replace only the state: each of N = 100 neurons carries $\mathbf{z}_{i,t} \in \mathbb{R}^D$, with D = 1,000.

The spike remains one bit. The state that produces it is 1,000-dimensional.

02 The model

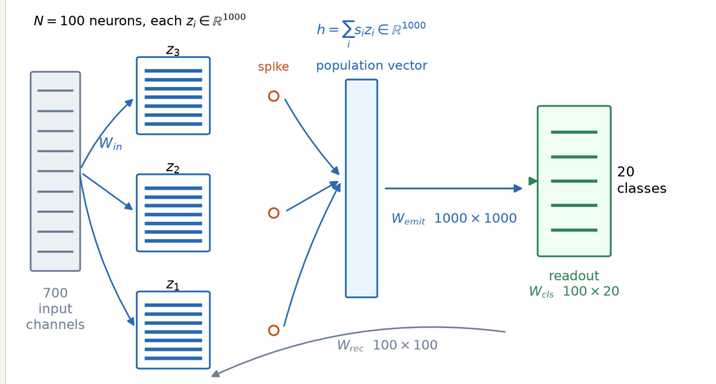


Figure 1. Schematic: input projection, vector states, shared emission and count readout. The rank-64 mixing runs inside each state (Section 05).

All coordinates of $\mathbf{z}_i$ participate in the dynamics. The model is not a scalar neuron with a vector appended to its input.

03 Where the vector enters

1 $a_t = x_t W_{in}, \quad d^{in}_{i,t} = g_i \odot a_t$

Input projection: 1,400 × 1,000. The gate $g_i \in \mathbb{R}^D$ selects coordinates.

2 $m_t = (z_t W_{down}) W_{up}$

Intrinsic mixing, rank R = 64: 128,000 parameters, not N × D × D.

3 $c_t = s_t W_{rec}, \quad h_t = (1/\sqrt{N})(\Sigma_j s_{j,t} z_{j,t}) W_{emit}$
$d^{rec}_t = c_t\, h_t$

Recurrent coupling and shared emission. A spike at neuron j gates the whole state vector z_j.

4 $z_{t+1} = (1 - \alpha_i) z_t + \alpha_i\, \tilde{z}_t$

Leaky vector state. $\alpha_i = \sigma(\text{raw}_i)$ is learned; the reported initialisation is α ≈ 0.1.

5 $o_{i,t} = \langle z_{i,t}, w_{out,i} \rangle + \theta_i - w_{ad} b_{i,t}$
$s_{i,t} = H(o_{i,t})$

The spike is a learned hyperplane through the whole state, not a scalar membrane potential.

The readout is unchanged: counts over time, then a linear classifier. Nothing about the spike is vector-valued — only the state behind it.

04 Parameter budget

Input + delay line	**1,400,000**
Emission matrix	**1,000,000**
Rank-64 mixing (×2 stages)	**256,000**
Recurrent, gate, bias, readout, z_0, classifier	**412,000**
Scalars and delays	**1,201**
Total	**3,069,201**

05 The full update

$$d_t = g_i \odot a_t + c_t h_t + \beta_i$$
$$m_t = (z_t W_{down}) W_{up}$$
$$m2_t = (z_t W_{down2}) W_{up2}$$
$$\tilde{z}_t = \tanh(\, m_t + \lambda \tanh(\, m2_t + d_t\,)\,)$$
$$z_{t+1} = (1 - \alpha_i) z_t + \alpha_i \tilde{z}_t$$
$$o_{i,t} = \langle z_{i,t}, w_{out,i} \rangle + \theta_i - w_{ad} b_{i,t}$$
$$b_{i,t+1} = \rho b_{i,t} + s_{i,t}, \quad s_{i,t} = H(o_{i,t})$$

$\lambda = 0.1 + 1.9\,\sigma(\lambda_{raw})$, initialised at 1.0. The reported model is the two-stage path with adaptation; the adLIF current block is present in the config but inert under two-stage mixing, so it is not shown.

METHODS · PROTOCOL AND CONTROLS

06 Training

$$L = CE + 10^{-4} \cdot \text{mean}_i (r_i - 10\ \text{Hz})^2$$

SHD split	**8,156 train / 2,264 test**
Input	**700 channels · 2 ms bins**
Sequence	**T = 500 + 50-bin tail**
Optimiser	**Adam · 10^{-3} · cosine**
Batch / epochs	**96 / 30**
Precision	**fp16 state · fp32 masters**
Thresholds	**10 Hz calibration**

BPTT with 10 gradient-checkpoint chunks; gradients clipped at 1.0. Thresholds are calibrated for three fixed-point iterations and stay trainable.

07 Augmentation

Same-class blend
Two raw event lists of the same class are centre-of-mass aligned; each event is kept independently with p = 0.5, then merged and sorted. This is a mixup variant for event trains, applied before binning.

Event dropout
Each event is removed independently with p = 0.1. This is the only other augmentation active in the reported run. Both are training-only; validation and test events are byte-identical.

08 Controls

Scalar at D = 1	**isolates state width**
Capacity-matched scalar, N = 1,411	**isolates parameters**
Broadcast, E = 1,000	**isolates transmission width**
Width sweep, D = 4…1,000	**dose–response**

All controls use the same data, readout, optimiser and epoch budget. They are the reason the headline is not just a parameter-count story.

09 It learns SHD

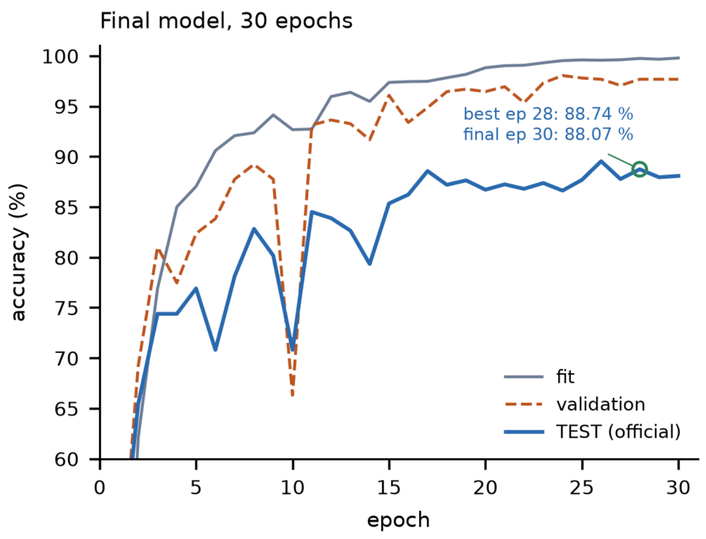


Figure 2. FIT, VAL and official TEST over 30 epochs. Best TEST 88.74%; final-epoch 88.07%.

88.07% TEST · 1,994 / 2,264
99.78% FIT · 7,340 trials
97.67% VAL · 816 trials
85.87% UNSEEN · speakers 4 & 5

10 Width, not parameters

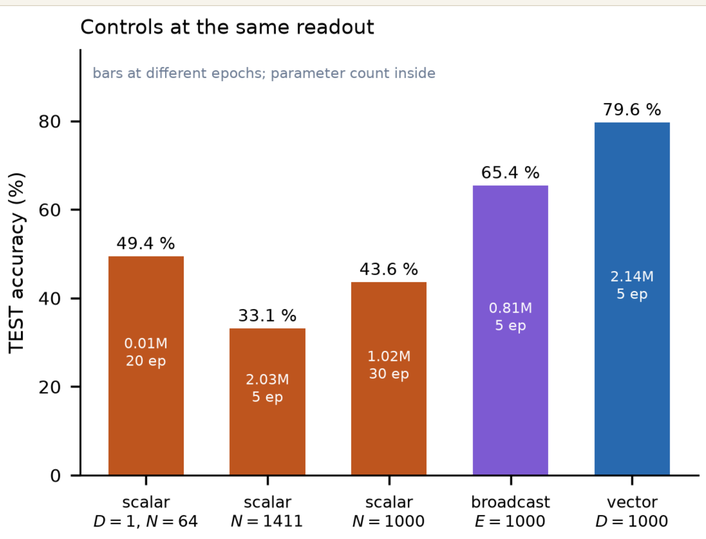


Figure 3. Scalar, capacity-matched scalar, broadcast and vector controls at the same readout.

D = 4 / 16 / 64	**57.20 / 63.74 / 68.60**
D = 256 / 1,000	**77.61 / 76.15**
Scalar, N = 100, 5 ep	**35.91**
Capacity-matched scalar	**33.13**
Broadcast, E = 1,000	**65.42**
Vector, N = 100, 5 ep	**79.64**

D = 4 → 256 buys +20.4 pp; D = 1,000 is not better than D = 256 at this budget. A 2M-parameter scalar reaches 33.13%.

RESULTS · WHAT THE STATE DOES

11 Sanity check: 1,000 scalar neurons

N = 1,000 · D = 1 · 30 EP
43.6%
TEST 987 / 2,264
FIT 49.6% · VAL 42.7%
1.02M params · 34 min

N = 100 · D = 1,000 · 5 EP
76.7%
TEST 1,736 / 2,264
FIT 96.4% · VAL 93.5%
2.14M params · 31 min

49.6% on FIT is the tell: the 1,000-neuron scalar does not fit the training split, so this is not a generalisation gap. In the same half-hour the vector gets 5 passes instead of 30 and still gains 33 points of TEST.

Not parameter-matched: 1.02M scalar vs 2.14M vector. The scalar spends 1M weights on a full recurrent matrix. The 30-epoch vector reaches 88.1% TEST / 99.8% FIT.

12 Probes

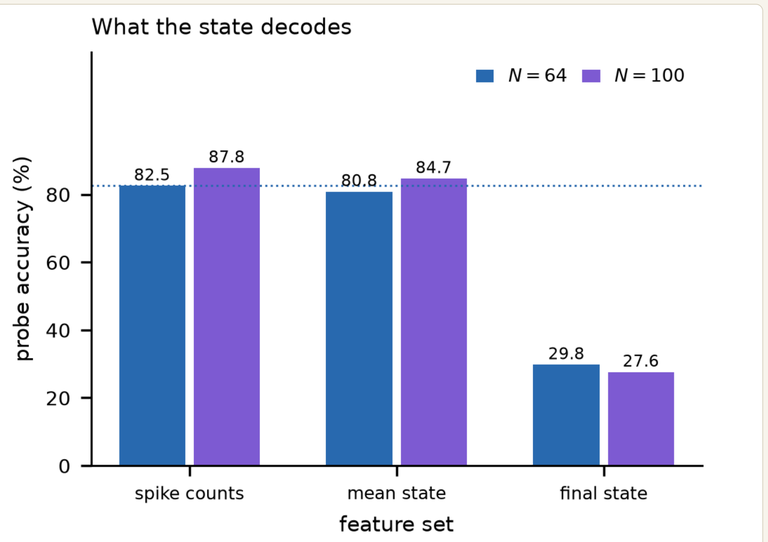


Figure 4. Linear probes on frozen states. Spike counts 87.81%; trial-mean state 84.67%; final state 27.61%.

The code is in the trajectory, not the endpoint. The population mean uses only **7.49 effective dimensions** out of 100,000.

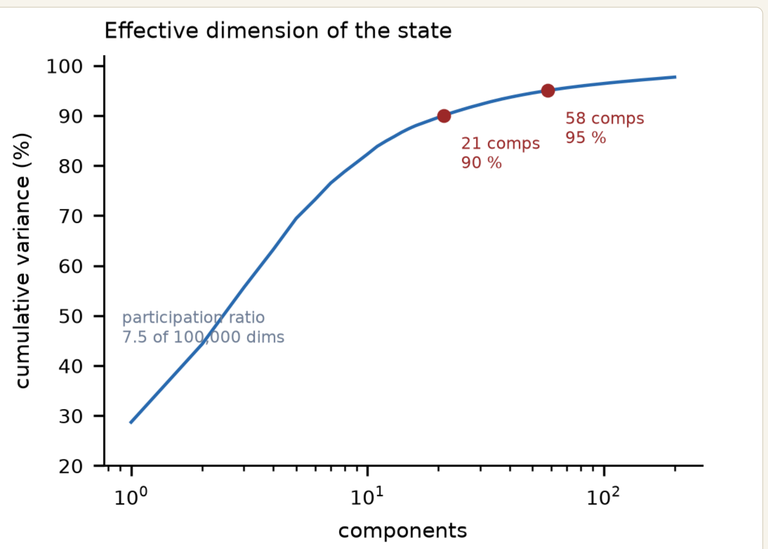


Figure 5. Effective dimension of the population state: 7.49 of 100,000; 21 components to 90%, 58 to 95%.

13 A distributed code

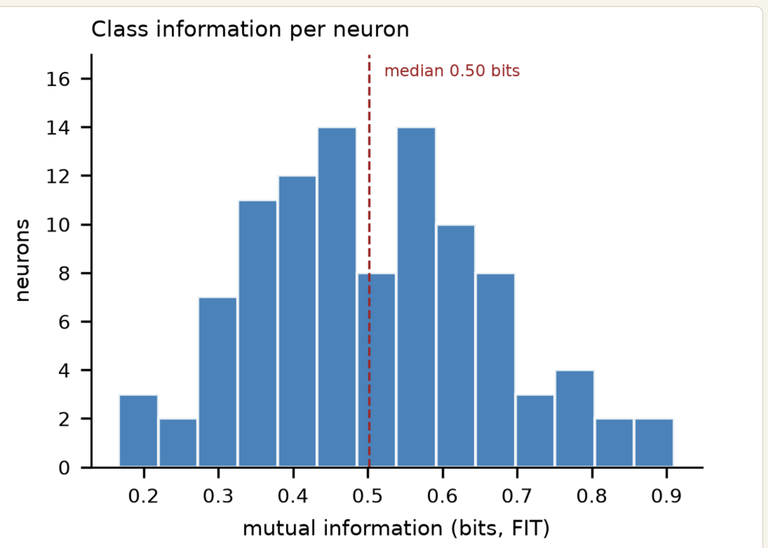


Figure 6. Mutual information per neuron: median 0.502 bits; 91 of 100 neurons informative.

Silencing one neuron costs **0.30 ± 0.63 pp** on average. No single unit is necessary: the code is redundant and distributed, which is what a vector state predicts and a dedicated scalar detector does not.

14 Limitations

- Single seed; no seed variance.
- Blocks and augmentations selected on 5-epoch TEST.
- No 30-epoch N = 100 all-off baseline.
- VAL is speaker-overlapping; the held-out pilot overstates.
- Scalar controls are untuned; the TEST set was read repeatedly.
- The 5-epoch vector is the earlier 2.14M configuration, not the final 3.07M model.

TAKEAWAY

Width of the state, not neuron count, is the variable that buys computation on SHD.

100 vector neurons reach 88.07%. A 1,000-neuron scalar does not even fit the training split.

15 What comes next

Design sketches, not measured results

Smaller footprint

$$W_{emit} \approx D + U V^T$$

A rank-16 emitter is about 33k parameters instead of 1M; factorising the input projection brings 1.4M down to about 38k.

Per-neuron embedding

$$g_i = W_g e_i, \quad \beta_i = W_\beta e_i, \quad w_{out,i} = W_o e_i$$

With $e_i \in \mathbb{R}^8$, the per-neuron heads cost about 8.8k for 1,000-D states, compared with 100k.

What would falsify it

If the SHD gain disappears at matched parameter budget, or under 30-epoch drop-one ablations, the claim is about parameters and training, not state width.

Provenance. Final: N = 100, D = 1,000, 30 epochs, seed 0, 3,069,201 parameters, official TEST n = 2,264. Controls: 5 epochs unless noted. Sanity check: N = 1,000, D = 1, 30 epochs, 1,024,703 parameters. All accuracy figures are on the official SHD test split; FIT and VAL are shown to separate fitting from generalisation.

Code and figures. [repository URL] · figure panels from the manuscript. Architecture and equations follow the reported configuration in *neuron_as_vector*.

Key references. Cramer et al. (2022), *IEEE TNNLS* 33(7):2744–2757, doi:10.1109/TNNLS.2020.3044364. Neftci et al. (2019), *IEEE Signal Processing Magazine*, doi:10.1109/MSP.2019.2931595. Zenke & Ganguli (2018), *Neural Computation* 30(6):1514–1541, doi:10.1162/neco_a_01086. Gu et al. (2021), *IJCAI*, doi:10.24963/ijcai.2021/97.